

Residue:-

If z_0 is a singular pt. of $f(z)$ and Laurent's expansion of $f(z)$ about $z = z_0$ is

$$f(z) = \sum_{n=0}^{\infty} a_n (z-z_0)^n + \underbrace{\frac{b_1}{z-z_0}}_{\text{residue}} + \frac{b_2}{(z-z_0)^2} + \dots + \frac{b_n}{(z-z_0)^n} + \dots$$

then b_1 ie. Coeff. of $\frac{1}{z-z_0}$ is called residue of $f(z)$ at $z = z_0$

Calculations of Residue at Poles:-

(i) If z_0 is a simple pole then,

Residue of $f(z)$ (at $z = z_0$)

$$= \lim_{z \rightarrow z_0} (z - z_0) f(z)$$

(ii) If z_0 is a pole of order n , then

Residue of $f(z)$ (at $z = z_0$)

$$= \frac{1}{(n-1)!} \lim_{z \rightarrow z_0} \frac{d^{n-1}}{dz^{n-1}} \left[(z - z_0)^n f(z) \right]$$

Q. Determine the poles of the function $f(z) = \frac{z^2}{(z-1)^2(z+2)}$ and also find the residue at each pole.

→ We have,

$$f(z) = \frac{z^2}{(z-1)^2(z+2)}$$

$$= \lim_{z \rightarrow 1} \frac{d}{dz} \left[\frac{z^2}{z+2} \right]$$

$$= \lim_{z \rightarrow 1} \frac{(z+2) \frac{d}{dz}(z^2) - z^2 \frac{d}{dz}(z+2)}{(z+2)^2}$$

Here, $z = 1$ is a pole of order 2.

& $z = -2$ is a pole of order 1.

Now, Residue of $f(z)$ (at $z = 1$)

$$= \frac{1}{(n-1)!} \lim_{z \rightarrow 1} \frac{d^{n-1}}{dz^{n-1}} \left[(z-1)^2 f(z) \right]$$

$$= \frac{1}{1!} \lim_{z \rightarrow 1} \frac{d}{dz} \left[\cancel{(z-1)^2} \times \frac{z^2}{\cancel{(z-1)^2}(z+2)} \right]$$

$$= \lim_{z \rightarrow 1} \frac{(z+2)(2z) - z^2(1)}{(z+2)^2}$$

$$= \frac{(1+2)(2) - (1)^2}{(1+2)^2}$$

$$= \frac{6-1}{9} = \frac{5}{9}$$

Q. Determine the poles of the function $f(z) = \frac{z^2}{(z-1)^2(z+2)}$ and also find the residue at each pole.

→ Residue of $f(z)$ (at $z = -2$)

$$= \lim_{z \rightarrow -2} (z+2) f(z)$$

$$= \lim_{z \rightarrow -2} \cancel{(z+2)} \times \frac{z^2}{(z-1)^2 \cancel{(z+2)}}$$

$$= \lim_{z \rightarrow -2} \frac{z^2}{(z-1)^2}$$

$$= \frac{(-2)^2}{(-2-1)^2} = \frac{4}{(-3)^2} = \frac{4}{9}$$

$$\text{Res.} = \lim_{z \rightarrow z_0} (z-z_0) f(z)$$

Q. Determine the order of a pole & find the corresponding residue. $f(z) = \frac{z^2 + 2}{z - 1}$

→ We have,

$$f(z) = \frac{z^2 + 2}{z - 1}$$

Clearly, $z = 1$ is a simple pole of $f(z)$.

Residue of $f(z)$ (at $z = 1$)

$$= \lim_{z \rightarrow 1} (z - 1) f(z)$$

$$= \lim_{z \rightarrow 1} (\cancel{z-1}) \times \frac{(z^2 + 2)}{(\cancel{z-1})}$$

$$= \lim_{z \rightarrow 1} z^2 + 2$$

$$= (1)^2 + 2$$

$$= 1 + 2$$

$$= 3$$

$$\text{Res} = \lim_{z \rightarrow z_0} (z - z_0) f(z)$$

Q. Determine the order of a pole & find the corresponding residue.

$$f(z) = \left(\frac{z}{2z+1} \right)^2 \quad \left| = \lim_{z \rightarrow -\frac{1}{2}} \frac{d}{dz} \left[\frac{(2z+1)^2}{4} \times \frac{z^2}{(2z+1)^2} \right] \right.$$

→ We have,

$$f(z) = \frac{z^2}{(2z+1)^2}$$

$$= \lim_{z \rightarrow -\frac{1}{2}} \frac{1}{4} \frac{d}{dz} (z^2)$$

Clearly, $z = -\frac{1}{2}$ is a pole of order 2.

$$= \lim_{z \rightarrow -\frac{1}{2}} \frac{1}{4} \times 2z$$

$$= \lim_{z \rightarrow -\frac{1}{2}} \frac{z}{2}$$

Residue of $f(z)$ (at $z = -\frac{1}{2}$)

$$= \frac{1}{1!} \lim_{z \rightarrow -\frac{1}{2}} \frac{d}{dz} \left[(z + \frac{1}{2})^2 f(z) \right]$$

$$= \lim_{z \rightarrow -\frac{1}{2}} \frac{d}{dz} \left[\left(\frac{2z+1}{2} \right)^2 \times \frac{z^2}{(2z+1)^2} \right]$$

$$= \frac{-\frac{1}{2}}{2} = -\frac{1}{4}$$

Q. Determine the order of pole and find the corresponding residue. $f(z) = \frac{e^z}{z^2 + \pi^2}$

→ We have

$$f(z) = \frac{e^z}{z^2 + \pi^2}$$
$$= \frac{e^z}{z^2 - (-\pi^2)}$$
$$= \frac{e^z}{z^2 - i^2 \pi^2}$$
$$= \frac{e^z}{(z + i\pi)(z - i\pi)}$$

Clearly $z = \pi i$ & $z = -i\pi$ are simple poles of $f(z)$.

Residue of $f(z)$ (at $z = \pi i$)

$$= \lim_{z \rightarrow \pi i} (z - \pi i) f(z)$$
$$= \lim_{z \rightarrow \pi i} (z - \cancel{\pi i}) \times \frac{e^z}{(\cancel{z - \pi i})(z + \pi i)}$$
$$= \lim_{z \rightarrow \pi i} \frac{e^z}{z + \pi i} \quad (e^{i\theta} = \cos\theta + i\sin\theta)$$
$$= \frac{e^{\pi i}}{\pi i + \pi i} = \frac{\cos\pi + i\sin\pi}{2\pi i} = \frac{-1 + 0}{2\pi i} = \frac{-1}{2\pi i}$$

Q. Determine the order of pole and find the corresponding residue. $f(z) = \frac{e^z}{z^2 + \pi^2}$

→ Residue of $f(z)$ (at $z = -\pi i$)

$$= \lim_{z \rightarrow -\pi i} (z + \pi i) f(z)$$

$$= \lim_{z \rightarrow -\pi i} (z + \pi i) \times \frac{e^z}{(z + \pi i)(z - \pi i)}$$

$$= \lim_{z \rightarrow -\pi i} \frac{e^z}{(z - \pi i)}$$

$$= \frac{e^{-\pi i}}{-\pi i - \pi i} = \frac{\cos \pi - i \sin \pi}{-2\pi i} = \frac{-1 - 0}{-2\pi i} = \frac{1}{2\pi i}$$

$$e^{-i\theta} = \cos \theta - i \sin \theta$$

$$\text{Res} = \lim_{z \rightarrow z_0} (z - z_0) f(z)$$